GEORGIA SOUTHERN UNIVERSITY

# Knowledge Graph Analytics for Privacy Policy Analysis

**Team Members:**
**Mehak Kaur[1] (Team Leader), Julie Allen[2]**

[1]mk12462@georgiasouthern.edu
[2]ja23916@georgiasouthern.edu

March 12, 2026

# Project Overview & Motivation

- Privacy policies govern how organizations collect, use, and share personal data
- Regulations are complex and organizations need a method to validate compliance
- Despite their legal importance, they are routinely written in dense, ambiguous language that users cannot meaningfully understand
- This project develops a Knowledge Graph (KG) framework for automated, comparative privacy policy analysis

# Why It Matters

- Users cannot meaningfully consent to what they do not understand
- Obfuscation is often deliberate; companies use vague language to limit user comprehension
- No existing tool allows users to compare privacy protections across platforms or industries
- Regulators lack visibility into industry-wide patterns and outliers

# Scope of the Review

- Covers automated analysis of privacy policies using NLP, machine learning, and knowledge graphs
- Reviews three core methodological streams: KG construction, structural analysis, and legal/linguistic analysis
- Situates our comparative framework relative to prior single-policy approaches
- Spans policies across social media, e-commerce, healthcare, finance, and entertainment sectors

GEORGIA SOUTHERN UNIVERSITY

# Research Questions

**RQ1:** How does privacy policy compliance vary across the financial, healthcare, and IT sectors, and what sector-specific patterns emerge?

**RQ2:** What are the practical implications of these compliance differences for customers in terms of data protection, user rights, and transparency?

**RQ3:** Can structural analysis of privacy policy knowledge graphs reveal meaningful industry clusters and compliance archetypes?

**RQ4:** Can predictive models identify privacy policies that deviate from regulatory expectations or industry norms?

GEORGIA SOUTHERN UNIVERSITY

# Literature Review Summary

## Major Themes

- **KG Construction**
  Automated extraction of policy entities and relations into graph structures
- **Policy Obfuscation**
  Deliberate use of vague language to obscure data practices from users
- **User Navigation**
  NLP tools designed to help non-experts extract meaning from legal text
- **Structural Complexity**
  Reveal hidden patterns in policy complexity using ML techniques

# Deliberately confusing language in terms of service and privacy policy agreements - (Yerby et al., 2021)

- **Research Question**: Is confusing language in privacy policies accidental or deliberate?
- **Methodology**: Legal and qualitative analysis of policy language; no computational methods used
- **Key Findings**: Obfuscation is intentional — companies exploit the absence of readability regulations to hide data practices; proposes federal reform modelled on consumer protection law
- **Strengths**: Uniquely explains the *why* behind the problem; provides legal grounding for automated solutions
- **Limitations**: No computational analysis; does not scale; does not propose a technical solution

# PoliGraph: Automated Privacy Policy Analysis using Knowledge Graphs - (Cui et al., 2023)

- **Research Question**: Can knowledge graphs automatically extract and structure privacy policy text into machine-readable, queryable representations?
- **Methodology**: Builds a KG from policy text using NLP - entities and relationships extracted as triples
- **Key Findings**: KGs successfully capture policy structure; policies become searchable and comparable at the sentence level
- **Strengths**: Fully automated; scalable to many policies; formal representation enables downstream querying
- **Limitations**: Single policy focus; no cross-sector comparison; no compliance or regulatory assessment

# Navigating Privacy Policies with NLP and Graph Mining: Advancements in User-Centric Legal Document Analysis - (Barrett et al., 2025)

- **Research Question**: How can NLP and knowledge graphs help ordinary users navigate complex privacy policy documents?
- **Methodology**: NLP pipeline combined with graph mining to extract actionable insights from policy text; user-centric design
- **Key Findings**: Structured graph representations significantly improve user ability to extract relevant information from dense legal text
- **Strengths**: User-facing focus; bridges the gap between legal text and everyday comprehension; practical tool design
- **Limitations**: Single policy analysis only; does not compare across platforms or sectors; no compliance measurement

# Unveiling Privacy Policy Complexity: An Exploratory Study Using Graph Mining, Machine Learning, and Natural Language Processing - (Ramasamy et al., 2025)

- **Research Question**: What hidden complexity patterns exist within privacy policies, and how can they be visualized?
- **Methodology**: Combines graph mining, machine learning, and NLP; uses dimensionality reduction and clustering to reveal structural patterns
- **Key Findings**: Privacy policies contain distinct complexity clusters; ML can reveal structural patterns invisible to manual reading; visualisation improves transparency
- **Strengths**: Closest to cross-policy analysis; strong use of ML; reveals patterns at scale
- **Limitations**: Does not address regulatory compliance; no sector-specific framing; does not answer what patterns mean for users

# KG for Automated Data Compliance (Joshi et al., 2020)

- **Research Question**: Can knowledge graphs automatically extract and structure privacy policy text into machine-readable, queryable representations?
- **Methodology**: Develops a KG utilizing OWL/RDF frameworks to define class hierarchies and NLP techniques for unstructured data extraction
- **Key Findings**: Machine processable RDF Graph of privacy policy text. Threat mapping to controls
- **Strengths**: Scalable; multi-policy assessment; Compliance assessment
- **Limitations**: Not all policies assessed; limited cross-sector comparison

# NLP Analysis of Web Privacy Policies (Belcheva et al., 2023)

- **Research Question**: What are privacy policy trends across websites (sectors) and over time? Is confusing language in privacy policies accidental or deliberate?
- **Methodology**: Longitudinal text analysis using NLP, text mining, and ML techniques. New Dale-Chall readability formula, taxonomy-based vague term counting, and fine-tuned BERT for vagueness classification. RegEx for positive phrases.
- **Key Findings**: Less readable, increased vagueness, yet more reassuring language
- **Strengths**: Large dataset with multiple sectors. Multiple measures of comprehensibility
- **Limitations**: Dataset is prior to 2020, and policies have changed

# PrivComp-KG for Compliance Verification (Garza et al., 2024)

- **Research Question**: How can LLMs and semantic web technologies be applied to align privacy policies with GDPR? How to identify compliance gaps without fine-tuning?
- **Methodology**: LLM with Retrieval-Augmented Generation (RAG) for extracting and aligning privacy policy sections with GDPR articles. Knowledge Graph (PrivComp-KG) population with RDF/OWL for storage
- **Key Findings**: Privacy compliance tool (PrivComp-KG)
- **Strengths**: 0.9 correctness; no model retraining required
- **Limitations**: Not all policies assessed. No real-time updates for laws. Dataset aging

# GraphDPR: PP Analysis with KGs and Topics (Chowdhury et al., 2026)

- **Research Question:** How can KG based semantic analysis improve understanding and comparison of complex privacy policies and compliance auditing for corporations?
- **Methodology:** NLP extracts entity–category–data triples, normalized with Sentence-BERT, structured as graphs in Neo4j, and analyzed using LDA topic modeling.
- **Key Findings:** Graph structures identify thematic patterns and relationships across privacy policies and organizations
- **Strengths:** Multi-policy analysis and cross-domain comparison
- **Limitations:** this could be expanded to other domains

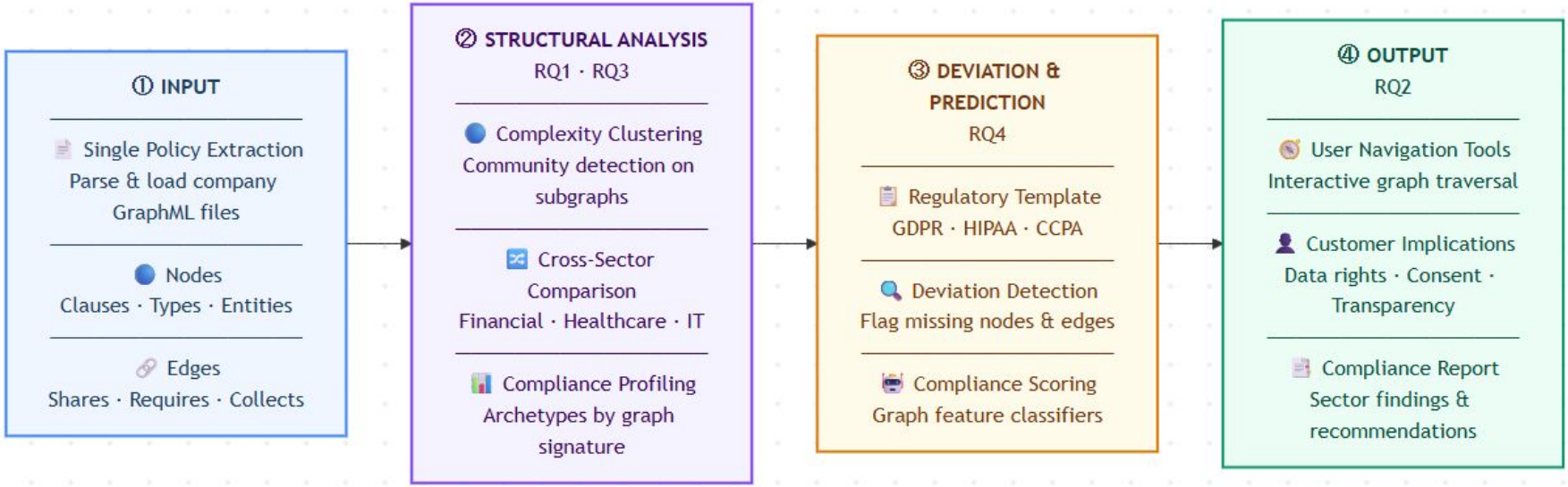
GEORGIA SOUTHERN UNIVERSITY

# Research Gaps

| What already exists | What's Missing - Our Solution |
|---|---|
| Single policy extraction | Cross-sector comparison |
| Why obfuscation happens | Compliance profiling |
| User navigation tools | Customer implications |
| Complexity clustering | Deviation detection |

GEORGIA SOUTHERN UNIVERSITY

# Proposed Framework



# Summary and Conclusion

**Problem**: Privacy policies and compliance regulations are often long and complex, and organizations and consumers face challenges as a result.

**Our Research Focus:** improve transparency for consumers and compliance for organizations

KGs can identify data about privacy policies and compliance regulations to identify:

- Privacy policy differences across sectors
- Industry clusters
- Deviations from regulations
- Implications for users

GEORGIA SOUTHERN UNIVERSITY

# References


[1] S. Barrett, V. Ramasamy, G. Dorai, and B. Boswell, “Navigating privacy policies with nlp and graph mining: Advancements in user-centric legal document analysis,” in SoutheastCon 2025, pp. 1013–1022, 2025.

[2] V. Ramasamy, S. Barett, G. Dorai, and J. Zumbach, “Unveiling privacy policy complexity: An exploratory study using graph mining, machine learning, and natural language processing,” in 2025 6th International Conference on Artificial Intelligence, Robotics and Control (AIRC), pp. 514–520, 2025.

[3] J. Yerby and I. Vaughn, “Deliberately confusing language in terms of service and privacy policy agreements,” Issues in information systems., vol. 23, no. 2, pp. 138–149, 2022.

[4] Z. Munn, J. Peters, Micah D. C. Stern, C. Tufanaru, A. McArthur, and E. Aromataris, “Systematic review or scoping review? guidance for authors when choosing between a systematic or scoping review approach,” BMC Medical Research Methodology, vol. 18, no. 1, p. 143, 2018.

[5] H. Cui, R. Trimananda, A. Markopoulou, and S. Jordan, “PoliGraph:Automated privacy policy analysis using knowledge graphs,” in 32nd USENIX Security Symposium (USENIX Security 23), (Anaheim, CA), pp. 1037–1054, USENIX Association, Aug. 2023.

[6] K. P. Joshi, L. Elluri, and A. Nagar, “An integrated knowledge graph to automate cloud data compliance,” IEEE Access, vol. 8, pp. 148541–148555, 2020.

GEORGIA SOUTHERN UNIVERSITY

# References (cont.)


[7] V. Belcheva, T. Ermakova, and B. Fabian, “Understanding website privacy policies—a longitudinal analysis using natural language processing,”Information, vol. 14, no. 11, 2023.

[8] L. Garza, L. Elluri, A. Piplai, A. Kotal, D. Gupta, and A. Joshi, “Privcomp-kg: Leveraging kg and llm for compliance verification,” in 2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA), pp. 97–106, 2024.

[9] H. Chowdhury, M. I. Morsalin, R. S. Azade, and V. Ramasamy, “Graphdpr: A privacy policy analysis framework using knowledge graphs and topic modeling.,” in Social Networks Analysis and Mining - 17th International Conference, ASONAM 2025, Proceedings, vol. 16322, pp. pp.422–430, 2026.

GEORGIA SOUTHERN UNIVERSITY

# Thank You!